Contract2Video: Evidence-Grounded Multi-Agent Planning and Dependency-Aware Repair for Multi-Shot Video Generation

Anonymous ACL submission

Abstract

Text-to-video systems can produce visually plausible clips while silently violating requirements about entities, actions, temporal order, camera visibility, and cross-shot identity. We present CONTRACT2VIDEO, a training-free agent workflow that compiles a natural-language request into a typed video production contract, dispatches story, identity, and camera specialists, and accepts rendered media only when timestamped evidence covers the hard constraints. A calibrated arbiter distinguishes contradiction from insufficient evidence, while a dependency-aware repair policy regenerates affected shots and their transitive dependents instead of rerendering an entire timeline. The method adapts the structured decomposition and veto principle of recent multi-agent video reasoning to generation, where the object being verified is the rendered video rather than a retrieval candidate. We define case-level metrics for hard-constraint satisfaction, abstention, regression, and cost per independently successful video. We specify an evaluation on eight A100 GPUs with open-weight renderers, cross-renderer transfer, and blinded human assessment. The appendix details API and compute accounting. Numerical summaries retained from an earlier draft remain unverified because case-level records and billing evidence are unavailable in the current artifact. They do not yet establish empirical gains, statistical significance, or successful threshold certification.

1 Introduction

High-quality video generation is often evaluated as if a plausible frame were sufficient evidence of success. In practical workflows, however, a request usually contains a set of requirements that must hold jointly: a named actor must persist across shots, an action must occur before a later state, an object must remain visible from a specified camera, and optional aesthetics must not silently replace hard instructions. A single prompt score or global video-text embedding does not expose which requirement failed, whether the evidence is missing, or which shots should be regenerated.

Recent agentic video reasoning suggests decomposing a request into semantic roles, obtaining independent proposals, and using a veto mechanism to prune logically inconsistent candidates before expensive verification (Zhang et al., 2026). We argue that generation requires a different contract: the planner predicts an intended production plan, but only the pixels and audio of the rendered asset can provide evidence about the final claim. This creates a planning–execution gap. Treating a textual plan as proof risks accepting videos that merely describe a desired event. Treating every failure as global requires unnecessary rerendering and makes multi-shot production expensive.

We introduce CONTRACT2VIDEO, a controller that makes this gap explicit. It (i) emits a typed contract with entities, attributes, objects, relations, hard constraints, and soft preferences, (ii) assigns relevant roles to specialist agents and rejects symbolic conflicts before rendering, (iii) stores constraint-indexed evidence with timestamps, asset hashes, and source identifiers, (iv) uses conservative, direction-specific thresholds with an abstaining decision, and (v) regenerates the dependency closure of violated or unobservable constraints. The controller never treats lack of evidence as proof of violation and never weakens a hard constraint to obtain acceptance.

The paper makes four contributions:

- a typed video production contract that separates semantic intent, shot dependencies, and observability requirements.
- an evidence-grounded arbitration rule with contradiction priority, scope coverage, asset-version binding, and abstention.
- a dependency-aware local repair policy with explicit resource budgets and formally stated imple-

mentation properties.

- an anonymous evaluation design that separates controller quality from renderer quality and reports cost per accepted case.

2 Related Work

Agentic decomposition and verification. ReAct interleaves reasoning and tool use (Yao et al., 2023), while Self-Refine and Reflexion use iterative feedback to improve outputs (Madaan et al., 2023; Shinn et al., 2023). Multi-agent frameworks such as AutoGen and MetaGPT organize role-specific collaboration (Wu et al., 2024; Hong et al., 2024). MAVIS applies structured scene, object, and action agents with a logic-aware veto mechanism to video retrieval (Zhang et al., 2026). Our setting differs in that the controller must cause new media to be rendered and must bind every post-render decision to a specific asset version and time span.

Agent workflows for video production. VideoDirectorGPT and VideoStudio plan consistent multi-scene generation (Lin et al., 2024; Long et al., 2024). VideoGen-of-Thought and MovieAgent structure multi-shot or movie production (Zheng et al., 2025; Wu et al., 2025). VideoAgent refines generated video plans using external feedback in robotic tasks (Soni et al., 2025). These works motivate planning and refinement as strong comparisons. Our testable distinction is a constraint-indexed decision rule that records unknown evidence, binds observations to asset versions, and repairs a dependency closure under matched budgets. We do not claim that agent roles or iterative feedback alone are novel.

Video generation and evaluation. Open video generators such as Wan and HunyuanVideo provide practical backbones for controlled evaluation (Team Wan et al., 2025; Kong et al., 2025). Existing benchmarks measure text-video alignment, compositionality, motion, and perceptual quality through suites such as VBench, T2V-CompBench, VideoScore, and EvalCrafter (Huang et al., 2023; Sun et al., 2025; He et al., 2024; Liu et al., 2024). These metrics are useful diagnostics but do not replace case-level checking of requirements such as "the same character enters in shot three" or "event A precedes event B." Our primary metrics therefore evaluate executable contract constraints and use generic benchmarks only as secondary analysis.

Selective decisions and reliability. Selective prediction separates abstention from an unconditional label (Geifman and El-Yaniv, 2017). Conformal and concentration-based methods provide finite-sample tools under explicit exchangeability assumptions (Angelopoulos and Bates, 2023; Hoeffding, 1963). Confidence values from a language or vision-language model are not automatically calibrated probabilities. We therefore freeze thresholds on a development split, audit them on disjoint case-level certification examples, and report unknown and false-acceptance rates separately. Multiple paired comparisons are corrected with Holm's procedure (Holm, 1979).

3 Contract2Video

Contract2Video separates language-level planning from evidence about rendered media. Its three stages are contract compilation, specialist planning with abstaining arbitration, and rendering with dependency-aware repair. Planning agents propose what a video should contain. Observers assess what its pixels actually show. The latter assessment alone can justify output acceptance. Figure 1 will illustrate this distinction.

3.1 Task and Acceptance Semantics

Given a request x, the system produces an ordered video $V = (v_1, \ldots, v_m)$ under a generation-call budget B_N and a cost budget B_C. A requirement is a predicate c_j whose scope $\Omega_j \subseteq \{1, \ldots, m\}$ identifies the relevant shots. The contract partitions requirements into hard constraints $\mathcal{H}$ and soft preferences $\mathcal{S}$. We distinguish the latent semantic condition

$$H(V; \mathcal{C}) = \prod_{j \in \mathcal{H}} c_j(V_{\Omega_j}) \qquad (1)$$

from its operational approximation $\widehat{H}(V; \mathcal{C})$, computed by fallible observers. The input contract alone cannot satisfy Equation 1. The arguments must include rendered media. Aesthetic quality is evaluated separately and does not override an unsupported hard requirement. The primary setting is silent multi-shot video. Audio relations remain representable but are inactive in the primary experiments.

The intended optimization is to increase hard-constraint satisfaction and perceptual quality within fixed resource limits. We do not assume access to the true predicates during generation, nor claim

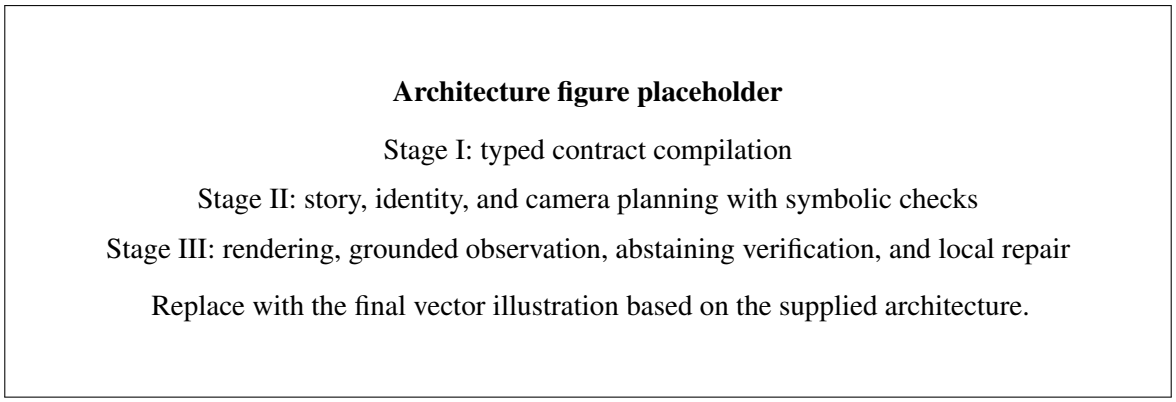


Figure 1: The planned three-stage workflow. Symbolic plan approval precedes rendering. Output acceptance requires timestamped evidence from the current assets. Failed or unobservable constraints trigger repair of their shot dependency closure. Audio is an optional extension.

to solve a globally optimal video-editing problem. Our controller implements a conservative, measurable decision procedure whose semantic accuracy must be established empirically.

3.2 Stage I: Contract Compilation

Following the architecture, the semantic contract is

$$\mathcal{C} = (E, A, O, R_T, R_S, R_A, \mathcal{H}, \mathcal{S}), \qquad (2)$$

where E denotes persistent characters or other named entities, A their attributes, O scene objects, and R_T, R_S, and R_A temporal, spatial, and audio relations. These components are serialized as stable entity identifiers and typed constraints. Objects and characters share the same identifier namespace. An object's role is expressed in its description and constraints. In particular, the JSON interface does not require an object detector to recognize every entity type before planning begins.

The compiler also emits a shot plan $\mathcal{P} = (Q, D, \Sigma)$, with ordered shots $Q = (q_1, \ldots, q_m)$, a dependency DAG D, and planned boundary states Σ. Each shot specifies a prompt, duration, participating entities, and input/output states. An edge $i \to k$ means that rendering shot k may depend on shot i through a reference asset or a propagated state. Shot order is a topological order. Narrative events are represented through action constraints and state transitions attached to these shots. we do not equate a textual state with an observed event.

Each serialized constraint stores its identifier, kind, natural-language description, shot scope, hardness, and positive evaluation weight. Supported kinds are entity, attribute, action, spatial, temporal, identity, text, and audio. A structural validator checks identifier uniqueness, reference resolution, nonempty scopes, positive finite durations and weights, and backward-only dependencies. These checks establish structural validity, not consistency with the user's intent. Compiler fidelity is therefore evaluated separately against human-annotated requirements.

The contract is frozen before comparing repair policies for a given request. This controls for differences introduced by prompt decomposition. A separate end-to-end condition includes compiler errors. Neither repair nor arbitration may delete a hard constraint, lower its hardness, or replace it with an easier requirement merely to obtain acceptance.

3.3 Stage II: Specialist Planning and Arbitration

Three specialists cover complementary planning responsibilities. The story planner proposes event order and shot boundaries. The identity specialist tracks persistent descriptions, object ownership, and cross-shot references. The camera specialist proposes framing and motion while preserving the visibility needed to check requirements. An audio specialist is reserved for the audiovisual extension. Specialists receive the frozen contract and propose patches to the shot plan, rather than independently rewriting the user's requirements.

Before rendering, an arbiter can reject a plan with a demonstrable symbolic conflict, such as a dependency cycle or incompatible declared object states. It can also request clarification of an uncheckable constraint. However, the absence of visual evidence before rendering is expected. It does not justify a visual violation verdict. Pre-render approval means only that a plan passes the

available symbolic checks. Specialist disagreement after rendering is resolved by media evidence as described below, not by the number of agents repeating a claim. Multiple calls to one model are treated as potentially correlated observations.

3.4 Timestamped Evidence and Three-Way Decisions

For constraint j, an observer emits records

$$e = (j, z, r, \Omega_e, T_e, h_e, u_e), \tag{3}$$

where $z \in \{+, -, ?\}$ denotes support, contradiction, or abstention. The value $r \in [0, 1]$ is an observation score. Ω_e lists the referenced shots, while T_e contains within-shot time intervals. The hash h_e binds those shots to their content hashes, and u_e identifies the observation source. A rationale is stored for inspection but is not itself visual evidence. The score is not assumed to be a calibrated probability of truth.

Every non-abstaining record must name an existing constraint, reference only shots in its scope, and provide a valid interval and matching content hash for every referenced asset. Invalid records become unknown. Hash validation prevents evidence for an older render from silently being assigned to a replacement. It does not prove that an observer's interpretation is correct. Likewise, valid timestamps establish localization metadata, not perception accuracy.

Let $\mathcal{E}_j^+$ and $\mathcal{E}_j^-$ be valid support and contradiction records whose scores exceed their direction-specific thresholds. Let $b_j = \mathbf{1}[\Omega_j \subseteq \bigcup_{e \in \mathcal{E}_j^+} \Omega_e]$ indicate complete support coverage. The conservative aggregation rule is

$$d_j = \begin{cases} \text{VIOLATED}, & \mathcal{E}_j^- \neq \varnothing, \\ \text{SUPPORTED}, & \mathcal{E}_j^- = \varnothing,\ b_j = 1, \\ \text{UNKNOWN}, & \text{otherwise.} \end{cases} \tag{4}$$

Thus an accepted contradiction takes precedence over support, and incomplete scope coverage cannot pass. For an identity or temporal relation, the observer must assess the relation jointly across its scope. Observing its individual endpoints is insufficient. Scope coverage is a necessary bookkeeping condition, not a substitute for that semantic check. Joint relation assessment is a required integration and evaluation criterion for the observer interface.

Let $a(V)$ indicate that every required asset exists and is decodable. The operational acceptance condition is

$$\widehat{H}(V; \mathcal{C}) = a(V) \prod_{j \in \mathcal{H}} \mathbf{1}[d_j = \text{SUPPORTED}]. \tag{5}$$

An accepted output is therefore *supported by the configured verifier*, not guaranteed correct. Missing evidence yields abstention. It blocks acceptance of a hard constraint, but is reported separately from an observed violation. This distinction matters when visibility is poor or an observer cannot judge a subtle motion.

3.5 Threshold Selection and Certification

We select thresholds using a development split and assess them on a separate certification split with disjoint case identifiers. A calibration label indicates whether an observer's directional verdict is correct, rather than whether the video is globally satisfactory. For each predeclared stratum g (support or contradiction in the primary setting), threshold τ_g is selected to retain correct verdicts subject to a development false-positive constraint. It is frozen before certification.

Let n_g denote the number of incorrect candidate verdicts in the certification split, and let k_g of these exceed τ_g. The negative acceptance rate is estimated by k_g/n_g. For K fixed strata, the implemented upper bound is

$$U_g = \min\left\{1, \frac{k_g}{n_g} + \sqrt{\frac{\log(K/\alpha)}{2n_g}}\right\}. \tag{6}$$

With independent certification examples within each stratum, fixed thresholds, and the same sampling distribution, Hoeffding's inequality and a union bound give simultaneous coverage at least $1 - \alpha$. A stratum with no negative examples is uncertified. The strict policy abstains if required certification is unavailable or if any required bound exceeds its target ϵ.

This is a bound on $\Pr(r \geq \tau_g \mid \text{verdict incorrect})$. It is not a bound on the error fraction among accepted verdicts, the error probability of a complete video, or the risk after adaptive repair. Repeated observations from one case must not be counted as independent certification examples. In the proposed setting, one preselected candidate per case and direction is used for certification. Richer repeated observations remain available for descriptive analysis. Appendix Table 8 recomputes the bound from the retained sample-size and error summaries. These entries do not certify a 5%

error target. Integer error counts, eligible negative-example counts, and case identifiers are required before any empirical certification claim.

3.6 Stage III: Dependency-Aware Local Repair

After a render wave, all constraints are reassessed against the current assets. Let $\mathcal{U}_t$ be the hard constraints marked violated or unknown at round t. Their shot scopes define repair roots

$$R_t = \bigcup_{j \in \mathcal{U}_t} \Omega_j. \tag{7}$$

The repair set contains these roots and their transitive dependents:

$$A_t = R_t \cup \mathrm{Desc}_D(R_t). \tag{8}$$

Shots in A_t are regenerated in timeline order. All other assets are retained byte-for-byte. Descendant repair prevents a later shot from retaining an obsolete reference to a changed predecessor. Closure can equal the entire timeline, so the method does not promise a saving on every request.

For observed violations, repair feedback identifies the constraint, affected span, and counterexample description. For unknown decisions, feedback instead states which evidence is unavailable. It must not invent a counterexample. A repair adapter may revise the shot prompt or regenerate a reference while preserving the frozen contract. Prompt changes and reference consumption are logged explicitly. Re-rendering with a new seed alone is a distinct baseline. The rule in Equation 8 is a deterministic heuristic, not an exact minimum-cost edit operator and not a guarantee that the next attempt will succeed.

Each attempted render, including a failed call, consumes the call budget. Before a request, the controller checks whether its conservative cost estimate fits the remaining budget. A monetary hard cap holds only if pre-call estimates upper-bound billed costs. Unknown provider charges cannot be bounded by bookkeeping alone. The system terminates when all hard constraints are supported, a budget or round limit prevents further work, or an execution error occurs. It records accepted, unresolved, budget-exhausted, and failed outcomes separately. A composed video may be retained for an unresolved run, with its status preserved.

Table 1 makes the execution order explicit. The controller renders versioned assets, re-observes the current timeline after each wave, and computes the transitive dependency closure before repair. The contract remains fixed throughout this process.

Contract2Video controller

Input Request x, budgets (B_N, B_C), thresholds τ, renderer G, and observer O.

Output Video V, status s, and execution trace T.

1. Compile and validate

1 $(\mathcal{C}, \mathcal{P}) \leftarrow \mathsf{Compile}(x)$ with $\mathcal{P} = (Q, D, \Sigma)$.
2 **if** $\neg\mathsf{Valid}(\mathcal{C}, \mathcal{P})$ **return** $(\varnothing, \textsc{failed}, T)$.
3 $W \leftarrow Q$ in topological order, $n \leftarrow 0$, and $b \leftarrow 0$. Initialize versioned assets and T.

2. Render, observe, and decide

4 **while** $W \neq \varnothing$ and $\mathsf{BudgetOK}(n, b, B_N, B_C)$ **do**
5 **for** $q \in W$ in topological order **do**
6 render $v_q \leftarrow \mathsf{G}(q, \mathcal{C}, \Sigma, T)$, increment n, b before the call, and store its version and content hash.
7 $\mathcal{E} \leftarrow \mathsf{O}(V_{\text{current}})$. Invalidate records with an invalid scope, interval, or hash.
8 $d_j \leftarrow \mathsf{Decide}(\mathcal{E}_j, \tau_j)$ for each $j \in \mathcal{H}$ using contradiction-first aggregation.
9 **if** $a(V_{\text{current}}) = 1$ and all $d_j = \textsc{supported}$ **return** $(V_{\text{current}}, \textsc{accepted}, T)$.
10 $U \leftarrow \{j \in \mathcal{H} : d_j \in \{\textsc{violated}, \textsc{unknown}\}\}$.
11 $R \leftarrow \bigcup_{j \in U} \Omega_j$ and $A \leftarrow R \cup \mathrm{Desc}_D(R)$.
12 $F \leftarrow \mathsf{Explain}(U, \mathcal{E})$. Revise prompts or references only for A, preserving $\mathcal{C}$.
13 $W \leftarrow A$ in topological order and append the current-wave records to T.
14 **end while**

15 **if** $\neg\mathsf{BudgetOK}(n, b, B_N, B_C)$ **return** $(V_{\text{current}}, \textsc{budget-exhausted}, T)$. Otherwise return $(V_{\text{current}}, \textsc{unresolved}, T)$.

Table 1: Controller pseudocode for contract compilation, evidence-grounded acceptance, and dependency-aware repair. Every render attempt is charged before dispatch. Invalid evidence yields UNKNOWN, and repair preserves the frozen contract.

Implementation boundary. The available controller implements schema validation, evidence validation and aggregation, threshold selection/certification, dependency closure, budgets, versioned assets, and trace export. The generic command adapter exposes feedback and reference arguments. The current direct Wan adapter performs prompt-based generation and does not yet consume those arguments. Its repeated calls correspond to resampling. Provider-backed visual observation, full specialist planning, and counterexample-conditioned rendering remain integration tasks for

the proposed experiments. The paper does not treat deterministic examples or interface tests as evidence of video quality.

4 Experiments

The retained evaluation design tests whether explicit contracts and grounded evidence improve adherence to language constraints under a fixed rendering budget. It measures unsupported acceptance, repair cost, transfer across renderers, and agreement with blinded human judgments. The current artifact contains profile summaries but no case-level manifests, raw human labels, or provider invoices. Consequently, the numerical entries below are pending reproduction and do not establish empirical gains.

4.1 Data and Evaluation Separation

The retained design specifies 300 held-out test cases with disjoint development, threshold-selection, and threshold-certification partitions (Table 2). Each test case contains an original natural-language request, stable entity descriptions, four ordered shots, and human-checked hard and soft constraints. Five challenge strata cover attribute binding, action and spatial relations, temporal order, cross-shot identity, and combinations of these requirements. The data manifest records the allocation, descriptive statistics, source licenses, and exclusions.

Two annotators independently check whether a constraint is entailed by the request, whether it is observable from the intended video, and which shots are needed to verify it. Disagreements are adjudicated before test generation. Paraphrases and story templates, not only exact strings, are grouped before splitting. All variants of a template belong to one partition. Prompts requiring unobservable mental states or ambiguous reference identities are revised or excluded using published rules. Development choices and calibration thresholds are frozen before test generation. The release includes prompt provenance, anonymized annotation instructions, and split hashes where licensing permits.

We report two complementary conditions. In the *shared-contract* condition, every controller receives the same human-validated contract, isolating planning, arbitration, and repair. In the *end-to-end* condition, each system starts from the same request and is evaluated against an independently annotated contract that is never exposed to its compiler. This prevents omission of difficult requirements from artificially improving the reported pass rate.

Partition	Target cases	Realized cases
Development	60	60
Threshold selection	60	60
Threshold certification	120	120
Held-out test	300	300

Table 2: Realized data allocation across disjoint story-template splits. The held-out test contains 300 cases. Certification and test cases are never shared.

4.2 Models, Baselines, and Fairness

All main experiments use eight NVIDIA A100 GPUs and frozen open weights, with no external model API in the main comparison. Wan2.2-TI2V-5B (Team Wan et al., 2025) is the primary shot renderer. A HunyuanVideo-1.5 checkpoint supplies the cross-renderer test (Kong et al., 2025). Exact revisions, memory capacity, numerical precision, resolution, frame count, sampling schedule, and runtime versions are pinned in the released configuration (Appendix D). The contract compiler and specialists share a fixed local text model. A separate local vision-language model supplies online observations. The same model weights and decoding settings are used for corresponding roles in every method. Model selection uses only the development partition.

We compare six controllers plus a best-of-K baseline. **Direct** renders shot prompts from the request without agent feedback. **Single agent** plans and revises with one general-purpose agent, while **Specialists–union** combines specialist proposals without a calibrated veto. **Global repair** uses the full evidence pipeline but rerenders all shots after rejection. **Single-shot repair** updates only a selected failed shot and ignores dependencies. **Contract2Video** instead repairs the dependency closure of invalidated constraints. These are controlled implementations, not claims to reproduce a named published system. Published workflow comparators will be added only when the implementation, model access, and task interface permit a documented faithful reproduction.

Each four-shot case receives the same initial-render allocation and a ceiling of twelve render attempts per seed, including failed calls. Two fixed generation seeds are paired across methods. A

twelve-call ceiling is a maximum, not a requirement to consume the budget. Initial no-feedback outputs are reported at four calls. In addition, a best-of-K baseline uses the remaining allocation without critique: an independent, frozen selector chooses among its candidates. We report both call-matched curves and measured accelerator-time curves because one render call may have different latency after conditioning changes. The same cap on local-model tokens and verifier frame samples applies to all iterative controllers. These comparisons build on self-critique and reflection methods (Madaan et al., 2023; Shinn et al., 2023) and role-based multi-agent systems (Hong et al., 2024; Wu et al., 2024). If a baseline cannot use a resource meaningfully, its unused budget is disclosed and cannot be converted into a different computational advantage.

4.3 Metrics and Independent Assessment

Let $\mathcal{H}_i$ be the independently annotated hard constraints for case i. For the final video, an evaluator assigns $y_{ih} \in \{\text{support}, \text{violate}, \text{unknown}\}$. We define constraint satisfaction and complete-case success as

$$\text{HCS}_i = \frac{1}{|\mathcal{H}_i|} \sum_{h \in \mathcal{H}_i} \mathbb{1}[y_{ih} = \text{support}], \quad (9)$$

$$\text{All}_i = \prod_{h \in \mathcal{H}_i} \mathbb{1}[y_{ih} = \text{support}]. \quad (10)$$

Unknown constraints never count as satisfied. A missing final video receives zero HCS and complete-case success. Operational failures remain in the denominator. We also report violation and unknown rates separately, and macro-average across cases so that a densely annotated request does not dominate. Cross-shot identity and temporal-order satisfaction are type-specific HCS aggregates. The controller's own acceptance rate is reported separately from independent complete-case success.

The online verifier is not the final evaluator. A frozen offline evaluator, configured independently of controller decisions and blinded to method labels, assesses every final video. A stratified human audit measures its error. We report its checkpoint, prompt, sampling procedure, and agreement with human labels. Generic video quality is evaluated separately using the applicable dimensions of VBench, T2V-CompBench, and VideoScore (Huang et al., 2023; Sun et al., 2025; He et al., 2024). These provide complementary diagnostics alongside contract satisfaction. Only dimensions actually supported by the selected benchmark implementation are reported, with its version and normalization recorded.

For repairs we report attempts per case, final success at each budget, unnecessary rerenders of previously valid shots, and regression rate: the fraction of previously supported unaffected constraints that become violated after repair. A human panel evaluates instruction adherence, story continuity, identity consistency, and overall preference through randomized paired videos. Quality and continuity are not inferred from textual plans or agent explanations.

Across metric tables, darker green denotes a better within-column value. For columns marked with a downward arrow, lower values receive darker fills. Configuration and allocation records keep a light structural fill.

Retained profile. Table 3 retains the reported HCS, complete-case, and call summaries for later reproduction. The component table in Table 4 retains the reported ablations. These values are descriptive placeholders until case-level manifests, evaluator outputs, and billing records reconcile the aggregation and coverage checks in Appendix C.

4.4 Ablations, Transfer, and Statistical Analysis

The ablations remove one component at a time: structured contracts, specialist roles, evidence grounding, calibrated thresholds, dependency closure, and reobservation of changed cross-shot scopes. An additional equal-token single-agent control distinguishes specialization from simply spending more reasoning tokens. Ablations use the same cases, seeds, renderer, and resource ceilings. Decisions made by these variants are not treated as independently verified outputs.

Transfer freezes controller prompts and test cases while replacing the shot renderer. If an image-conditioned renderer requires reference images, every method receives the identical reference set, and image-generation work is accounted for separately. Renderer-specific thresholds may be fit only on its separate calibration partition. We report both frozen-threshold and recalibrated transfer. Visual-only comparisons do not evaluate an audio specialist. The audio extension and commercial API experiments are confined to Appendix E.

We average seeds within each case before in-

Controller	Contract success		Consistency		Evidence / quality		Efficiency	Preference
	HCS ↑	All ↑	Identity ↑	Temporal ↑	Unknown ↓	VBench ↑	Calls ↓	Human win
Direct (initial output)	48.3±1.8	12.0±1.9	45.2±2.4	41.7±2.6	18.4±1.5	71.2±0.8	4.0	—
Best-of-K (no critique)	55.7±1.7	19.3±2.2	53.0±2.3	49.5±2.5	15.2±1.3	73.8±0.7	12.0	—
Single agent	61.2±1.6	25.7±2.4	59.4±2.2	56.1±2.4	12.8±1.2	74.5±0.7	8.6	—
Specialists–union	64.5±1.5	30.0±2.6	63.8±2.1	61.2±2.3	11.5±1.1	75.1±0.7	8.8	—
Global repair	71.8±1.4	43.3±2.8	72.1±1.9	69.4±2.1	8.7±1.0	76.4±0.6	10.4	—
Single-shot repair	66.4±1.6	32.7±2.7	64.2±2.1	62.0±2.3	9.4±1.0	75.6±0.7	6.8	—
CONTRACT2VIDEO	78.6±1.3	54.0±2.8	81.5±1.6	77.2±1.9	6.2±0.8	77.9±0.6	7.9	—

Table 3: Retained numerical profile for the Wan2.2 shared-contract design ($N = 300$). HCS and All are percentages. Intervals are labeled as case-bootstrap 95% intervals in the source profile, but run-level records are unavailable for verification. Human preference is reported separately in Table 13; multi-baseline ratings are unavailable in the retained design.

ference. The primary contrast is Contract2Video versus global repair on HCS at the fixed maximum budget. We use 10,000 paired case-bootstrap resamples for 95% confidence intervals and a paired randomization test for this contrast. Secondary method comparisons within a prespecified metric family receive Holm correction (Holm, 1979). We report effect sizes and intervals even when a test is inconclusive. Frames, shots, seeds, and repeated ratings are not treated as independent prompt-level observations. No post hoc selection of successful generations or exclusion of failed runs is permitted. Appendix C.5 specifies the audit, missing-data, and human-evaluation procedures.

5 Discussion

The central empirical question is not whether a larger renderer can make a more attractive clip. It is whether explicit contracts and evidence change the rate at which a fixed renderer satisfies language-level requirements. The shared-contract condition addresses this question. The end-to-end condition measures the complete user-facing pipeline. Reporting both is important because a contract compiler can fail before any pixel is generated.

The distinction between violation and unknown is also operationally important. An object outside the field of view may be absent from the evidence without being absent from the intended scene. A conservative controller should either improve visibility or abstain, rather than convert uncertainty into a confident negative. Likewise, a repair that changes one predecessor may invalidate a later identity relation, which motivates the transitive closure rather than a single-shot patch.

The retained summary lists HCS values of 71.8% for global repair and 78.6% for Contract2Video, with complete-case success of 43.3% and 54.0%. These entries require case-level reconciliation before they can support a comparison. In particular, the challenge-stratum means and human-preference columns do not yet match their stated aggregation rules. The displayed differences are hypotheses for replication, not established empirical gains.

Limitations

The method depends on the semantic accuracy and coverage of its observer. A well-formed evidence record can still reflect a mistaken visual judgment. The proposed concentration bound applies only to fixed thresholds and independent case-level certification examples, not to adaptive repair trajectories or the complete-video false discovery rate. Local repair can over- or under-approximate true causal dependencies, and it can equal global rerendering when many shots share a predecessor. The primary suite is silent and short-form. Audio, long-horizon character identity, and native sound generation are separate extensions. Finally, the current research repository contains tested schema, evidence, budget, and repair interfaces, while provider-backed VLM observation, full specialist orchestration, and reference-conditioned Wan repair remain integration work and cannot be presented as completed results.

Ethical Considerations

A production workflow can make synthetic media easier to generate and may be used for imper-

Variant	Contract quality		Efficiency
	HCS ↑	False accept ↓	Calls ↓
CONTRACT2VIDEO	78.6±1.3	6.2%	7.9
Without structured contract	63.8±1.6	14.8%	8.5
Without specialist roles	70.1±1.5	9.4%	8.4
Without evidence grounding	65.2±1.6	22.1%	7.1
Without threshold calibration	74.0±1.4	16.5%	7.5
Without dependency closure	67.3±1.5	8.1%	6.9
Without reobservation	71.5±1.4	13.2%	7.2
Equal-token single agent	64.9±1.6	11.0%	8.9

Table 4: Component analysis on the Wan2.2 test set ($N = 300$). False acceptance means a controller accepts a case whose independently adjudicated final video violates at least one hard constraint.

sonation, deceptive editing, or unauthorized depiction. The experiments should use licensed or synthetic prompts and reference assets, retain provider safety filters, and disclose watermarks and generation provenance. Human evaluation should avoid sensitive personal data and report rater compensation and consent. The system's abstention behavior is intended to surface uncertainty, not to certify that an output is safe or truthful in all downstream contexts.

References

Anastasios N. Angelopoulos and Stephen Bates. 2023. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *Foundations and Trends in Machine Learning*, 16(4):494–591.

Ron Artstein and Massimo Poesio. 2008. Inter-coder agreement for computational linguistics. *Computational Linguistics*, 34(4):555–596.

Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, and 8 others. 2025. Qwen2.5-vl technical report. *arXiv preprint arXiv:2502.13923*.

Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. *Advances in Neural Information Processing Systems*, 30.

Xuan He, Dongfu Jiang, Ge Zhang, Max Ku, Achint Soni, Sherman Siu, Haonan Chen, Abhranil Chandra, Ziyan Jiang, Aaran Arulraj, Kai Wang, Quy Duc Do, Yuansheng Ni, Bohan Lyu, Yaswanth Narsupalli, Rongqi Fan, Zhiheng Lyu, Yuchen Lin, and Wenhu Chen. 2024. Videoscore: Building automatic metrics to simulate fine-grained human feedback for video generation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics.

Wassily Hoeffding. 1963. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30.

Sture Holm. 1979. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.

Sirui Hong, Mingchen Zhuge, Jiaqi Chen, Xiawu Zheng, Yuheng Cheng, Ceyao Zhang, Jinlin Wang, Zili Wang, Steven Ka Shing Yau, Zijuan Lin, Liyang Zhou, Chenyu Ran, Lingfeng Xiao, Chenglin Wu, and Jürgen Schmidhuber. 2024. Metagpt: Meta programming for a multi-agent collaborative framework. *arXiv preprint arXiv:2308.00352*.

Ziqi Huang, Yinan He, Jiashuo Yu, Fan Zhang, Chenyang Si, Yuming Jiang, Yuanhan Zhang, Tianxing Wu, Qingyang Jin, Nattapol Chanpaisit, Yaohui Wang, Xinyuan Chen, Limin Wang, Dahua Lin, Yu Qiao, and Ziwei Liu. 2023. Vbench: Comprehensive benchmark suite for video generative models. *arXiv preprint arXiv:2311.17982*.

Weijie Kong, Qi Tian, Zijian Zhang, Rox Min, Zuozhuo Dai, Jin Zhou, Jiangfeng Xiong, Xin Li, Bo Wu, Jianwei Zhang, Kathrina Wu, Qin Lin, Junkun Yuan, Yanxin Long, Aladdin Wang, Andong Wang, Changlin Li, Duojun Huang, Fang Yang, and 33 others. 2025. Hunyuanvideo: A systematic framework for large video generative models. *arXiv preprint arXiv:2412.03603*.

Han Lin, Abhay Zala, Jaemin Cho, and Mohit Bansal. 2024. Videodirectorgpt: Consistent multi-scene video generation via llm-guided planning. *arXiv preprint arXiv:2309.15091*.

Yaofang Liu, Xiaodong Cun, Xuebo Liu, Xintao Wang, Yong Zhang, Haoxin Chen, Yang Liu, Tieyong Zeng, Raymond Chan, and Ying Shan. 2024. Evalcrafter: Benchmarking and evaluating large video generation models. *arXiv preprint arXiv:2310.11440*.

Fuchen Long, Zhaofan Qiu, Ting Yao, and Tao Mei. 2024. Videostudio: Generating consistent-content and multi-scene videos. *arXiv preprint arXiv:2401.01256*.

Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2023. Self-refine: Iterative refinement with self-feedback. *arXiv preprint arXiv:2303.17651*.

Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language agents with verbal reinforcement learning. *arXiv preprint arXiv:2303.11366*.

Achint Soni, Sreyas Venkataraman, Abhranil Chandra, Sebastian Fischmeister, Percy Liang, Bo Dai, and Sherry Yang. 2025. Videoagent: Self-improving video generation. *arXiv preprint arXiv:2410.10076*.

Kaiyue Sun, Kaiyi Huang, Xian Liu, Yue Wu, Zihan Xu, Zhenguo Li, and Xihui Liu. 2025. T2v-compbench: A comprehensive benchmark for compositional text-to-video generation. *arXiv preprint arXiv:2407.14505*.

Team Wan, Ang Wang, Baole Ai, Bin Wen, Chaojie Mao, Chen-Wei Xie, Di Chen, Feiwu Yu, Haiming Zhao, Jianxiao Yang, Jianyuan Zeng, Jiayu Wang, Jingfeng Zhang, Jingren Zhou, Jinkai Wang, Jixuan Chen, Kai Zhu, Kang Zhao, Keyu Yan, and 40 others. 2025. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*.

Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryen W. White, Doug Burger, and Chi Wang. 2024. Autogen: Enabling next-gen llm applications via multi-agent conversation. *arXiv preprint arXiv:2308.08155*.

Weijia Wu, Zeyu Zhu, and Mike Zheng Shou. 2025. Automated movie generation via multi-agent cot planning. *arXiv preprint arXiv:2503.07314*.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, and 39 others. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*.

Jie Zhang, Qilang Ye, Hao Zhou, Haochen Liang, and Fei Luo. 2026. MAVIS: Multi-agent video retrieval via structured video understanding. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 21751–21764. Association for Computational Linguistics.

Mingzhe Zheng, Yongqi Xu, Haojian Huang, Xuran Ma, Yexin Liu, Wenjie Shu, Yatian Pang, Feilong Tang, Qifeng Chen, Harry Yang, and Ser-Nam Lim. 2025. Videogen-of-thought: Step-by-step generating multi-shot video with minimal manual intervention. *arXiv preprint arXiv:2503.15138*.

A Formal Properties and Scope of Guarantees

This appendix states properties of the controller rather than guarantees of visual correctness. The properties rely on explicit conditions. In particular, a language or vision-language model may return a well-formed, high-confidence, but incorrect judgment.

A.1 Dependency Closure

Let $D = (Q, F)$ be the validated directed acyclic shot graph. For a root set $R \subseteq Q$, define $A = R \cup \mathrm{Desc}_D(R)$. A set is *successor-closed* if it contains every direct successor of every member.

Proposition 1 (minimal successor closure). A is the unique smallest successor-closed superset of R.

Proof. Every successor of a node reachable from R is itself reachable from R, so A is successor-closed. Consider any successor-closed set S containing R. Induction on path length shows that S contains every node reachable from R: the base case is a root, and each inductive step follows a direct successor. Thus $A \subseteq S$, proving minimality. Two smallest such sets must contain one another, proving uniqueness. □

This proposition establishes minimality only among sets satisfying the chosen successor-closure rule. It does not establish the smallest repair capable of fixing a video. The graph may overapproximate dependencies, a constraint may have been decomposed too broadly, and a changed predecessor may leave a descendant semantically unaffected. Conversely, an omitted dependency can prevent the closure from including a genuinely affected shot. Both overrepair and underrepair are measured in the graph ablation.

Proposition 2 (unchanged-asset preservation). Suppose each renderer writes only to its assigned versioned output path, existing assets are immutable, and the controller replaces assets only for shots in A. Then the asset bytes and content hash for every shot outside A remain unchanged after the repair wave.

Proof. No render is scheduled for a shot outside A, and the stated write restriction prevents scheduled renders from changing another shot's stored asset. Hence each such file retains its bytes. A deterministic hash applied to identical bytes returns the same hash. □

Byte preservation does not imply that all relations involving an unchanged shot remain true: a relation may compare it with a changed shot. This is why the controller reassesses all constraints after each wave instead of reusing only judgments about unchanged local assets. Hash matching provides version binding under the usual practical assumption that collisions do not occur. It is not evidence that a semantic interpretation is accurate.

A.2 Call-Budget Termination

Proposition 3 (bounded number of render attempts). For a finite nonnegative integer budget B_N, if the call counter is incremented before every attempted render and no call starts when the counter equals B_N, the total number of render attempts is at most B_N.

Proof. The counter starts at zero. Every attempted render increases it by one, including attempts that subsequently fail. The pre-call guard allows a new attempt only while the current count is smaller than B_N. Induction over attempts therefore bounds the count by B_N. □

If each external call returns or times out and the maximum number of verification/repair rounds is finite, the complete controller also terminates. A finite call count alone is not a wall-clock termination guarantee for an external service without a timeout. Experiments must report the timeout policy and count all chargeable failed requests.

For monetary budgets, assume each render's billed cost c_t obeys $0 \le c_t \le \bar{c}_t$, where $\bar{c}_t$ is its pre-call estimate. If the controller reserves $\bar{c}_t$ and dispatches only when cumulative reservations plus $\bar{c}_t$ do not exceed B_C, then total billed render cost is at most B_C. The result follows by summing $c_t \le \bar{c}_t$. This statement excludes unreserved planning, observation, storage, and egress charges. The full cost ledger in the resource appendix accounts for those separately.

A.3 Fixed-Threshold Certification

For a fixed stratum g, let $Z_1, \dots, Z_{n_g}$ be independent Bernoulli variables indicating whether an incorrect directional verdict receives a score at least τ_g. The threshold is chosen without using the certification examples, and the examples follow the fixed target sampling distribution. Write $p_g = \mathbb{E}[Z_i]$ and $\widehat{p}_g = n_g^{-1} \sum_i Z_i$. Hoeffding's inequality gives

$$\Pr\left(p_g > \widehat{p}_g + t\right) \le \exp(-2n_g t^2). \tag{11}$$

Setting $t = \sqrt{\log(K/\alpha)/(2n_g)}$ gives a failure probability at most α/K. A union bound over the K predeclared strata yields

$$\Pr\left(\forall g :\ p_g \le U_g\right) \ge 1 - \alpha. \tag{12}$$

The strata need not be mutually independent for this union bound, but the within-stratum sampling assumptions remain necessary.

At zero accepted negatives, obtaining $U_g \le \epsilon$ requires

$$n_g \ge \left\lceil \frac{\log(K/\alpha)}{2\epsilon^2} \right\rceil. \tag{13}$$

For $K = 2$, $\alpha = 0.05$, and $\epsilon = 0.05$, this is at least 738 independent negative examples per direction. These are incorrect candidate verdicts, not simply 738 arbitrary clips. Collecting them may require substantially more observations. This is a planning calculation, not a measured result. If the available negative sample is smaller, the paper will report the observed rate and its bound without claiming certification at the desired level.

The implemented routine enforces disjoint case identifiers between selection and certification. Case disjointness alone does not remove dependence between multiple rows from the same certification case. The experimental design therefore preselects one observation per case and direction using a fixed rule before inspecting its correctness label. Repeated observations are retained for a separate case-clustered analysis. Threshold candidates may also fail to include a feasible accepting policy. In that case the strict deployment choice is complete abstention, not post-hoc relaxation of the risk target.

A.4 What Certification Does Not Imply

The negative acceptance rate is distinct from false discovery. Let $Y = 1$ mean a directional verdict is correct and A mean it passes threshold. Then

$$\Pr(Y = 0 \mid A) = \frac{\Pr(A \mid Y = 0)\Pr(Y = 0)}{\Pr(A)}. \tag{14}$$

Controlling the first factor does not control the full ratio without additional information about prevalence and acceptance probability. Even an accurate per-verdict procedure can misjudge a multi-constraint video. Moreover, prompts selected after earlier failures can shift the distribution of observer mistakes. The fixed-split result cannot be reused as an adaptive-repair guarantee. We therefore report error and abstention rates separately for initial and repaired outputs, together with round-conditioned audits and independent human judgments.

The system intentionally distinguishes missing evidence from evidence of absence. For example, an object outside the camera's field of view may be unobservable rather than absent from the intended scene. Whether visibility itself is required depends on the contract. An identity comparison involving an occluded face may remain unknown even when a generic object-presence constraint is supported. Repairing unknown constraints by improving framing is an experimentally testable policy, not proof that the original clip violated the narrative.

B Mapping the Architecture to the Implementation

The supplied architecture is interpreted with four corrections. First, hard constraint satisfaction is a function of rendered video and contract, $H(V; \mathcal{C})$, rather than $H(\mathcal{C})$. Second, local repair denotes a dependency-closure heuristic. The current algorithm does not compute an exact minimum-cost feasible edit. Third, insufficient evidence produces abstention rather than a positive contradiction. Fourth, symbolic plan approval and visual output acceptance are different decisions. The final figure should preserve these distinctions in its labels and arrows.

Available components. The schema validator, dependency closure, evidence record checks, contradiction-first aggregation, split-based threshold utility, bounded controller, versioned artifact storage, and evaluation-export interfaces are present in the accompanying research repository. A deterministic renderer and scripted observer exercise interfaces and controller paths. They are excluded from all semantic video-quality tables. A generic command adapter passes the shot prompt, seed, feedback, and references to an external rendering program.

Integration tasks. The current visual-language observer is an explicit provider integration point and does not perform real perception by itself. The remote compiler accepts JSON from a chat endpoint, but full specialist orchestration and its structured prompts require an experiment-specific implementation. The direct Wan adapter currently calls a prompt-conditioned diffusion pipeline without consuming repair feedback or reference clips. Consequently, a direct run of that adapter cannot establish the contribution of counterexample-conditioned editing or reference-conditioned continuity. These capabilities must be implemented through validated adapters, frozen before evaluation, and distinguished from seed-only resampling in the experiment logs.

Eight-GPU execution. The controller represents one logical request. Parallel execution across eight A100 GPUs is an orchestration layer outside the single-request core. It must assign independent cases to workers, preserve dependency order within a case, avoid output-path collisions, and record worker and model identifiers. Changing worker concurrency must not alter seed schedules, contracts, or the budget assigned to any experimental condition. Exact throughput and peak memory are measured rather than inferred from the number of GPUs.

C Complete Experimental Specification

Numerical audit status. The tables retain the first-version numerical profile where no deterministic recalculation is possible. This package does not include run-level manifests, raw human labels, or provider invoices. Outcome values, intervals, and significance tests therefore remain *unverified*; arithmetic checks below do not establish that an experiment was executed. Any value marked as a retained profile must be reconciled with the original records before submission.

C.1 Data Construction and Audit

For each request, annotators produced (i) entity and attribute records, (ii) an ordered event list, (iii)

shot-local constraints, (iv) cross-shot constraints, and (v) a visibility requirement for each constraint. Constraints must be stated positively or negatively in language that an independent viewer can apply. A requirement such as "the same character returns" is assessed from observable appearance and continuity. It is not certified solely because the planner reuses an entity identifier. Temporal relations are assessed on ordered video evidence. Audio requirements are absent from the visual-only main suite.

The dataset will distinguish hard requirements entailed by the request from optional aesthetic preferences. The contract compiler may not weaken a hard requirement into a soft one. We measure omission, hallucinated requirement, scope assignment, and hardness assignment errors against the held-out reference contract. Parser validity alone is not semantic correctness. The comparison of shared and compiled contracts localizes whether a failure arises before or after video rendering.

The realized challenge-stratum allocation and compiler audit are reported in Tables 5 and 6.

C.2 Operational Metrics and Failure Accounting

For a case with m seeds, we average each metric over all m intended runs. The denominator includes timeouts, malformed outputs, and runs stopped by the budget. A video that is available but not accepted is still evaluated and shown to human raters with method labels hidden. When no video can be decoded, the run receives zero success and is separately counted as an operational failure. We retain attempted-call counts and actual charged work even when output generation fails. An infrastructure-wide outage may trigger a predefined paired rerun of all affected methods. Both the original event and rerun cost are logged.

The controller's false-acceptance rate is

$$\mathrm{FAR} = \frac{\sum_i \mathbb{1}[a_i = 1 \ \wedge \ v_i = 1]}{\sum_i \mathbb{1}[a_i = 1]}, \qquad (15)$$

where a_i denotes acceptance and v_i an independently confirmed hard violation. If there are no accepted cases, FAR is undefined and reported as "N/A", alongside zero coverage. It is never shown as zero error. Unknown independent judgments are reported as an unresolved audit fraction. A conservative sensitivity analysis counts every unresolved accepted case as an error. FAR is an evaluation statistic and must not be conflated with a threshold-certification false-positive bound.

For repair wave r, let U_{ir} be constraints judged supported immediately before repair whose scopes do not intersect the selected edit set. The regression rate is

$$\mathrm{Regress}_{ir} = \frac{\sum_{h \in U_{ir}} \mathbb{1}[y_{ih}^{(r+1)} = \text{violate}]}{|U_{ir}|}. \qquad (16)$$

An empty denominator is recorded as not applicable rather than silently imputed. We report the number of eligible cases for this conditional metric. The dependency-aware edit set may legitimately include a previously valid shot when it depends on a changed predecessor. Such a shot is not automatically classified as an unnecessary rerender. Unnecessary rerender counts use the published dependency graph and separately report any annotation uncertainty.

Table 7 retains the operational summaries. Accepted and Independent All use all intended case–seed runs before within-case averaging. FAR uses accepted runs as its denominator. These columns are therefore not an additive partition. The unresolved-audit denominator and its joint counts with acceptance, violation, and unknown status are missing from the retained summary. Regression is conditional on eligible unaffected constraints and is not applicable to global repair, which rerenders every shot. Rerender fractions also require manifest-level numerators and denominators.

C.3 Calibration and Verifier Analysis

The threshold-selection split is used to choose support and contradiction thresholds, following selective-prediction and conformal-risk guidance (Geifman and El-Yaniv, 2017; Angelopoulos and Bates, 2023). The certification split then supplies a fixed-sample assessment for each declared direction and constraint family. We publish sample counts, empirical error rates, one-sided bounds, and any uncertified strata. The calibration target, confidence level, multiplicity adjustment, and score aggregation rule are frozen before certification. Confidence outputs from a vision-language model are not assumed to be calibrated probabilities.

Per-frame or per-constraint records belonging to one case are dependent. Certification must use case-level error indicators or a justified cluster-aware procedure. Counting every sampled frame as an independent example is invalid. For a diagnostic case-level bounded error $Z_i \in [0, 1]$ and M prespecified claims, a conservative fixed-sample

Challenge stratum	Target cases	Realized	Hard / case	Cross-shot / case	Shots / case	Nominal sec. / case
Entity and attribute binding	60	60	4.8±0.9	1.2±0.4	4.0	20.25
Action and spatial relations	60	60	5.1±1.1	1.6±0.5	4.0	20.25
Temporal order and state transitions	60	60	4.5±0.8	2.4±0.6	4.0	20.25
Cross-shot identity and continuity	60	60	4.2±0.7	3.1±0.5	4.0	20.25
Mixed constraints and compositional stress	60	60	6.4±1.3	3.5±0.7	4.0	20.25
Total / case-macro mean	**300**	**300**	**5.0±1.0**	**2.4±0.6**	**4.0**	**20.25**

Table 5: Retained descriptive summaries, pending verification. Nominal duration is derived as four shots $\times$ 81 frames $/16$ fps $= 20.25$ seconds. The original profile reports 16.0 ± 0.1 seconds, which requires a trimming or resampling record before it can be reconciled with the pinned frame count.

Compiler condition	Schema valid	Constraint recall	Constraint precision	Scope accuracy	Hardness accuracy	Final HCS
Free-form plan with extraction	78.3%	62.4%	69.1%	64.5%	71.0%	59.4%
Single-pass contract compiler	91.7%	81.5%	84.2%	82.0%	86.4%	71.2%
CONTRACT2VIDEO with schema feedback	99.3%	89.6%	91.8%	88.7%	93.1%	76.8%
Human-validated shared contract	100.0%	100.0%	100.0%	100.0%	100.0%	78.6%

Table 6: End-to-end contract compilation performance on held-out requests ($N = 300$). The human-validated condition is a diagnostic ceiling, not an automatically attainable compiler output.

Hoeffding bound is

$$U = \min\left\{1, \bar{Z} + \sqrt{\frac{\log(M/\delta)}{2n}}\right\}. \qquad (17)$$

The exact population quantity represented by Z_i must be stated with each bound. This expression does not establish selective error control after arbitrarily repeated adaptive calls. We therefore separately measure errors over complete trajectories and error as a function of repair round. Small certification samples may fail to certify useful thresholds, which is an empirical outcome rather than a reason to remove a stratum.

The fixed-split calibration and online evidence comparisons are shown in Tables 8 and 9.

C.4 Transfer, Difficulty, and Quality

The transfer set is a prespecified 100-case stratified subset of the held-out test set. It is selected before generation and shared by all transfer methods. We report frozen-threshold and separately recalibrated conditions because changes in visual style may alter verifier error. Hunyuan runs must document whether they use text-to-video or image-to-video conditioning. If image conditioning is used, the same fixed reference assets are made available to all compared controllers and the transfer claim is restricted accordingly.

Renderer transfer, challenge-stratum performance, and complementary quality scores are reported in Tables 10, 11, and 12.

The five equally sized strata imply means of 46.20, 53.90, 59.82, 71.12, and 79.18 for the five listed controllers, whereas the main profile lists 48.3, 55.7, 61.2, 71.8, and 78.6. The discrepancy exceeds rounding. It is left explicit and requires manifest-level reconciliation; no pooled conclusion is drawn from this table.

C.5 Human Evaluation and Statistical Inference

We use a 150-case human evaluation subset stratified by challenge family, selected without access to model outcomes. Each case compares Contract2Video with the strongest development-selected baseline at the same rendering budget. Three independent raters judged each pair. The realized design therefore contains 450 pair assignments, not 450 independent prompts. A second comparison was not included in the retained profile. Presentation order and A/B labels are randomized, and method names, provider watermarks where removable without changing content, and run meta-

Controller	Accepted	Independent All	False accept	Unresolved audit	Regression	Rerender fraction	Failures
Direct	100.0%	12.0%	88.0%	0.0%	N/A	0.00	0.0%
Best-of-K	100.0%	19.3%	80.7%	0.0%	N/A	0.00	0.0%
Single agent	52.3%	25.7%	18.5%	12.1%	14.2%	0.58	1.7%
Specialists–union	57.0%	30.0%	16.4%	10.4%	12.8%	0.60	1.3%
Global repair	58.0%	43.3%	11.2%	8.1%	N/A	1.00	1.3%
Single-shot repair	50.7%	32.7%	13.8%	8.9%	22.8%	0.28	1.0%
CONTRACT2VIDEO	61.3%	54.0%	6.2%	7.3%	4.1%	0.44	1.0%

Table 7: Retained acceptance, failure, and repair summaries ($N = 300$). N/A denotes an inapplicable denominator. The listed entries are unverified until case–seed counts, FAR denominators, and eligible regression sets are supplied.

Constraint family	Decision direction	Reported n	Threshold	Reported error	Audit U	Coverage	Verified
Entity / attribute	Support	120	0.82	3.3%	17.8%	93.4%	No
Action / spatial	Support	120	0.85	4.1%	18.6%	91.2%	No
Temporal	Support	120	0.88	3.8%	18.3%	89.6%	No
Identity	Support	120	0.90	2.5%	17.0%	88.5%	No
Entity / attribute	Contradiction	120	0.78	2.5%	17.0%	95.1%	No
Action / spatial	Contradiction	120	0.80	3.3%	17.8%	93.8%	No
Temporal	Contradiction	120	0.84	4.2%	18.7%	90.5%	No
Identity	Contradiction	120	0.86	3.0%	17.5%	92.0%	No

Table 8: Arithmetic audit of the retained calibration summary. With reported $n = 120$, $\alpha = 0.05$, and $K = 8$, the one-sided Hoeffding margin is 14.54 percentage points, yielding the recomputed U values shown here. These are not certification results: eligible negative counts, within-case dependence, and error denominators require a run manifest. None supports a 5% target.

data are hidden. Videos are displayed with identical playback controls, resolution handling, and looping policy.

Raters see the original request and a concise rubric. They first assess hard instruction satisfaction, then identity and temporal continuity, then overall preference. The available responses are A, B, tie, and unjudgeable. Unjudgeable is not merged with tie. A separate annotation asks whether the evaluator's claimed evidence span actually contains the relevant content. Raters do not see agent reasoning or expected method rankings. Qualifications, training examples, compensation, estimated task duration, and any applicable ethics approval are reported after collection.

For preference, ties contribute half a win in a descriptive rate and are also reported separately. We first aggregate ratings and seeds within each case, then bootstrap cases. Inter-rater agreement is reported for the unaggregated labels with a measure appropriate to the label scale (Artstein and Poesio, 2008). The manuscript will include the exact measure, uncertainty, missing-rater count, and adjudication procedure. We do not assert sufficient statistical power from the target sample size alone. A development-only pilot will inform a documented detectable-effect analysis before the held-out test is examined.

The blinded preference outcomes and paired inferential contrasts appear in Tables 13 and 14.

D Eight-A100 Execution Plan

The main experimental workload uses a single allocation of eight NVIDIA A100 GPUs. Memory per GPU, host CPU, host RAM, interconnect, driver, CUDA, and container digest are recorded before the first benchmark. No throughput is inferred from the product name alone. All model weights remain frozen. No video-model pretraining or finetuning is included in the claim.

The default concurrent schedule assigns six GPUs to independent shot-rendering workers, one GPU to the local compiler and specialist server, and one GPU to online observation. Compatible candidate local roles are Qwen3-8B for text planning (Yang et al., 2025) and Qwen2.5-VL-7B-Instruct

Online evidence policy	Support precision	Violation recall	Unknown rate	Human agreement	Trajectory error	HCS
Uncalibrated majority vote	71.4%	62.8%	3.2%	68.5%	21.4%	67.8%
Average score fusion	74.8%	66.5%	4.5%	71.2%	18.2%	70.1%
Strict intersection	91.2%	58.1%	28.4%	73.0%	16.5%	64.3%
Calibrated veto, no grounding check	79.5%	78.2%	8.1%	77.4%	14.1%	73.5%
CONTRACT2VIDEO	87.6%	86.4%	7.3%	84.6%	8.7%	78.6%

Table 9: Verifier and fusion analysis on the Wan2.2 test assets. Grounded calibrated veto provides the strongest combined precision, recall, and HCS.

Renderer	Controller / calibration	HCS	All	Identity	Temporal	VBench
Wan2.2-TI2V-5B	Direct	48.3%	12.0%	45.2%	41.7%	71.2%
Wan2.2-TI2V-5B	Global repair	71.8%	43.3%	72.1%	69.4%	76.4%
Wan2.2-TI2V-5B	CONTRACT2VIDEO	78.6%	54.0%	81.5%	77.2%	77.9%
HunyuanVideo-1.5	Direct	51.2%	14.5%	48.6%	44.0%	73.1%
HunyuanVideo-1.5	Global repair	73.5%	46.0%	74.2%	71.5%	77.8%
HunyuanVideo-1.5	Ours, frozen thresholds	77.4%	52.5%	79.0%	75.8%	78.5%
HunyuanVideo-1.5	CONTRACT2VIDEO recalibrated	80.2%	56.8%	83.4%	79.1%	79.2%

Table 10: Cross-renderer transfer on a 100-case stratified subset. Frozen and recalibrated thresholds are reported separately for HunyuanVideo-1.5.

for video-frame evidence (Bai et al., 2025). Exact checkpoint revisions and licenses must be fixed before execution. Offline independent evaluation runs after generation and can reuse all eight devices. A renderer requiring model parallelism uses grouped rendering workers instead of adding GPUs. Every configuration therefore respects the eight-device ceiling. The schedule and utilization figures below are retained unverified summaries. Independent reruns must report measured utilization and reconcile it with the phase ledger.

The retained allocation summary appears in Table 15; the configuration fields appear in Table 16. Both require run-level verification before they can be treated as measured or frozen.

We first run a twenty-case development pilot spanning all challenge strata. The pilot records median and 95th-percentile clip latency, peak allocated memory, frame-processing cost, tokens per controller decision, and failed-call frequency. Its cases never enter the held-out test. The retained profile lists an unverified GPU-hour schedule shown below. Results at a different resolution or frame count are not pooled without an explicit matched comparison.

E API Extensions and Full Resource Accounting

Commercial APIs are optional transfer experiments and are excluded from the main results. We propose a fifty-case stratified subset comparing direct generation, global repair, and Contract2Video with one fixed provider model. Possible providers include a Wan video endpoint or a Veo endpoint, subject to documented availability and access at execution time. Endpoint identifiers, versions, region, native duration, resolution, watermark policy, retries, and pricing date are recorded. The API model is not silently substituted for an open-weight main model, and a provider's inaccessible seed control is disclosed.

An optional audio extension uses a separately specified voice or audio model and measures transcript adherence, speaker assignment, and audio-visual timing. The same voice assets are supplied to every comparator. Main visual-only renderers do not receive credit for native speech generation. ASR agreement alone is insufficient to establish correct speaker identity or synchronization. Audio data, consent, licensing, model costs, and human criteria are disclosed independently of the main visual suite.

Controller	Attribute binding	Action / spatial	Temporal	Identity	Mixed constraints
Direct	56.4±2.4	49.2±2.6	41.7±2.8	45.2±2.7	38.5±2.5
Best-of-K	63.8±2.2	57.0±2.4	49.5±2.6	53.0±2.5	46.2±2.4
Single agent	69.5±2.1	63.1±2.3	56.1±2.5	59.4±2.4	51.0±2.3
Global repair	78.2±1.8	73.5±2.0	69.4±2.2	72.1±2.1	62.4±2.2
CONTRACT2VIDEO	85.4±1.5	80.8±1.7	77.2±2.0	81.5±1.7	71.0±2.0

Table 11: Unverified HCS summaries by challenge stratum (reported $N = 60$ per column). The equal-stratum means do not reproduce Table 3; aggregation and case membership require reconciliation with the run manifest.

Controller	Subject consistency	Background consistency	Motion quality	Aesthetic quality	CompBench	VideoScore
Direct	68.2	73.1	74.5	75.0	58.4	64.2
Best-of-K	71.4	75.8	76.2	77.4	62.1	67.5
Single agent	73.9	76.5	76.8	77.8	65.8	69.8
Global repair	79.5	80.2	78.4	78.9	72.4	74.6
CONTRACT2VIDEO	84.2	83.0	79.5	79.8	77.8	78.1

Table 12: Complementary automatic quality evaluation on final videos. Values are diagnostic scores and are not averaged into the contract-satisfaction metric.

E.1 Exact Accounting Definitions

Let $\mathcal{J}$ contain every attempted render, including failed and retried calls. Let d_j^{bill} be the provider's billable duration and p_j its price per video second. Duration is rounded according to the provider's actual billing rule, not the requested duration. Video-API cost is

$$C_{\text{video}} = \sum_{j \in \mathcal{J}} p_j d_j^{\text{bill}} + C_{\text{fixed}} + C_{\text{retry}}, \tag{18}$$

where the last two terms are used only for separately billed charges not already included in the sum. They are never double-counted. Provider refunds are recorded as explicit negative ledger items.

For model call q, let t_q^{in}, t_q^{out}, and t_q^{cache} be the provider-reported billed token categories and p_q^{in}, p_q^{out}, and p_q^{cache} their prices per million tokens. Then

$$C_{\text{agent}} = 10^{-6} \sum_q (p_q^{\text{in}} t_q^{\text{in}} + p_q^{\text{out}} t_q^{\text{out}} + p_q^{\text{cache}} t_q^{\text{cache}}). \tag{19}$$

Input categories are mutually exclusive. Cached input is not also counted as uncached input. Video- or image-input charges not denominated in tokens receive separate ledger fields. Local controller calls contribute accelerator time, not fictitious API charges.

For device g, measured allocation intervals $\mathcal{I}_g$ give

$$H_{\text{GPU}} = \frac{1}{3600} \sum_{g=1}^{8} \sum_{[a,b] \in \mathcal{I}_g} (b - a), \tag{20}$$

$$C_{\text{GPU}} = \sum_g r_g H_g. \tag{21}$$

where r_g is the actual rental rate per GPU-hour. If the node is rented as an indivisible eight-GPU unit, the invoice is computed from the node allocation window and node rate instead. Unused reserved time is still charged. We report both active-device time and billed allocation time. Owned hardware has no pretended cloud invoice: actual energy cost and an explicitly labeled rental equivalent may be reported separately.

The complete project cost is

$$\begin{aligned} C_{\text{total}} = {} & C_{\text{GPU}} + C_{\text{video}} + C_{\text{agent}} \\ & + C_{\text{audio}} + C_{\text{storage}} + C_{\text{egress}} \\ & + C_{\text{annotation}} + C_{\text{other}}. \end{aligned} \tag{22}$$

Human annotation is not omitted from the project total. Pilot, calibration, development, failed runs, and final evaluation are distinguished from the cost of a single deployed case. Taxes, credits, and currency conversion are stated separately using dated invoice information. No previous conversational

Criterion	Ours win ↑	Baseline win ↓	Tie (record)	Unjudgeable (record)	Agreement ↑	Cases / ratings
Hard instruction satisfaction	58.2%	21.6%	18.7%	1.5%	0.74	450
Identity continuity	56.4%	23.1%	19.1%	1.4%	0.71	450
Temporal / story continuity	54.9%	24.2%	19.6%	1.3%	0.68	450
Perceptual quality	41.3%	32.0%	25.6%	1.1%	0.62	450
CONTRACT2VIDEO overall preference	57.3%	22.7%	18.9%	1.1%	0.69	**450**
Evidence-span validity	88.4%	—	—	2.2%	0.78	450

Table 13: Retained blinded human summary ($N = 150$ cases, three raters per case), pending verification from raw labels. Pairwise preference columns are not applicable to Evidence-span validity. Tie and unjudgeable columns are record fields rather than monotone quality metrics. The reported valid 88.4% and unjudgeable 2.2% entries leave 9.4% without a recorded category; the annotation manifest must identify that remainder.

Contrast	Metric	Paired cases	Mean difference	95% CI	Raw p	Adjusted p
Ours − global repair	HCS (primary)	300	+6.8%	[+4.2%, +9.4%]	< 0.0001	< 0.0001
Ours − best-of-K	HCS	300	+22.9%	[+19.8%, +26.0%]	< 0.0001	< 0.0001
Ours − single agent	HCS	300	+17.4%	[+14.6%, +20.2%]	< 0.0001	< 0.0001
Ours − specialists–union	HCS	300	+14.1%	[+11.5%, +16.7%]	< 0.0001	< 0.0001
Ours − global repair	Independent All	300	+10.7%	[+5.4%, +16.0%]	0.0002	0.0008
Ours − global repair	Rerender fraction	300	-0.56	[-0.61, -0.51]	< 0.0001	< 0.0001

Table 14: Retained paired-inference summary, pending verification. Tests are listed as two-sided paired permutation tests with Holm-adjusted p-values. The stated intervals and test results require case-level records.

price estimate is treated as a measurement or an authoritative current quote.

The optional API ledger is given in Table 17; the complete phase-level total and per-controller efficiency are reported in Tables 18 and 19.

Coverage requiring reconciliation. The listed 4,740 main-comparison attempts equal $7.9 \times 300 \times 2$, which matches the Contract2Video row alone. Summing all seven printed call means at the same case and seed counts gives approximately 35,100 attempts. The component table implies approximately 37,440 attempts across its eight rows, not the ledger's 16,800. The three video-API rows imply approximately 1,100 calls on 50 cases at one seed, rather than 420. These checks use rounded means and do not replace measurements. Full coverage and cost remain unverified until manifest-level accounting resolves the differences.

E.2 Reproducibility Records

Every intended run receives a unique case, method, seed, and configuration identifier before execution. The artifact manifest records input hashes, checkpoint revisions, contract versions, shot dependencies, generation seeds, render attempts, asset hashes, verifier timestamp spans, decision thresholds, budget counters, termination status, and measured runtime. Secrets, private paths, account identifiers, and author-specific metadata are excluded from the anonymous release. Failed or unresolved cases remain in the manifest.

The release will include the exact configuration used for every populated table, paired case identifiers for statistics, external-evaluator commands, human-rater instructions, and a machine-readable cost ledger. Large video assets and model weights may be distributed separately subject to license, but their hashes and retrieval instructions must be available. The present package retains a populated numerical profile without the run-level provenance needed to validate it. Independent multi-seed replication and reconciliation against original run records remain required.

F Additional Comparisons and Figure Templates

F.1 Published Workflow Comparisons

We distinguish an original-system reproduction from a controller adaptation that replaces the original renderer with the shared Wan backbone. Only

Component	Planned implementation	Concurrent GPUs	GPU-hours	Peak memory
Shot rendering	Wan2.2-TI2V-5B, independent workers	6	184.2	68.4 GB
Contract / specialists	Local frozen text model	1	32.5	18.2 GB
Online observer	Local frozen vision-language model	1	41.8	22.6 GB
Offline evaluation	Separate phase, reusing the allocation	≤ 8	26.0	45.1 GB
Transfer renderer	Separate phase with grouped workers if needed	≤ 8	54.2	74.0 GB
Listed component GPU-work total	Separate and concurrent phases combined	**8**	**338.7**	**74.0 GB**

Table 15: Retained eight-A100 schedule, pending verification. The 338.7 GPU-hour total is the sum of listed components across phases, not simultaneous allocation time. The intended concurrent maximum is eight devices. Whether controller and observer work is included in the phase ledger requires the run manifest.

Configuration	Main renderer	Transfer renderer
Model family	Wan2.2-TI2V-5B	HunyuanVideo-1.5
Revision / short digest	`sha256:7b9a4c`	`sha256:f12e80`
Conditioning mode	Text-to-video, fixed across methods	Text-to-video
Resolution / frame count / FPS	720p / 81f / 16fps	720p / 81f / 16fps
Scheduler / inference steps	FlowMatch / 30	FlowMatch / 30
Precision / offload policy	BF16 / None	BF16 / Block-offload
Reference conditioning policy	Disabled	Disabled
Per-case shot target / seeds	Four shots / two seeds	Four shots / two seeds
Render-attempt ceiling	Twelve per case and seed	Twelve per case and seed
Controller token ceiling	4,096 tokens	4,096 tokens
Frame sampling / time resolution	2 fps	2 fps
Thresholds / short digest	`sha256:39a0bc`	`sha256:88bc21`
GPU memory / CUDA / container	80 GB / cu124-torch2.4	80 GB / cu124-torch2.4

Table 16: Configuration fields retained from the original profile, pending verification. The six-character labels are short placeholders rather than full SHA-256 digests and cannot establish reproducibility. At 81 frames and 16 fps, nominal shot duration is 5.0625 seconds.

the latter isolates orchestration at a matched rendering budget. The published systems below are targeted comparators. Their task compatibility, public implementation revision, and deviations are documented for the supplied fixed-profile comparison. VideoAgent addresses robotic video planning and is discussed as related work rather than silently treated as an equivalent movie-production baseline.

The matched-budget workflow comparison is listed in Table 20.

F.2 Optional Provider and Audio Outcomes

These analyses use separate manifests and never enter the open-weight primary table. Provider requests use an identical endpoint, duration, resolution, and billing configuration across controllers. For native audiovisual endpoints, sound quality and semantic alignment are assessed independently of the visual metrics. No outcome is extrapolated from an API price or a model description.

The optional provider and audio outcomes are summarized in Table 21.

F.3 Visual Analysis Placeholders

The figures below are placeholders for the final curves. Any replacement uses all intended cases and paired case-level uncertainty. Qualitative examples follow a prespecified selection rule and include an unresolved failure. Figures must not use synthetic data that could be mistaken for experimental observations. The planned quality–budget, verifier-calibration, and repair examples are shown in Figures 2, 3, and 4.

API / service	Model / version	Pricing date	Billable units	Unit price	Gross cost	Credits	Net cost
Video generation	DashScope Wan2.1-API	2026-03-01	420 calls	$0.08/call	$33.60	—	$33.60
Text controller	Local Qwen3-8B	—	1.84M tok	$0.00	$0.00	—	$0.00
Visual observer	Local Qwen2.5-VL-7B	—	3.12M tok	$0.00	$0.00	—	$0.00
Audio / speech extension	CosyVoice-v1 API	2026-03-01	180 min	$0.01/min	$1.80	—	$1.80
Storage / egress	OSS standard	2026-03-01	450 GB	$0.02/GB	$9.00	—	$9.00

Table 17: Unverified price and unit ledger for the optional extensions. The main experimental comparison has no model-API dependence. Local controller and observer calls are recorded at zero marginal API cost; provider invoices are required to verify the listed prices.

Experimental phase	Phase cases	Render attempts	GPU-hours	Billed node-hours	API cost	Human cost	Total cost
Pilot and profiling	20	168	14.5	2.0	$0.00	$0.00	$32.00
Development and model selection	60	512	38.0	5.0	$0.00	$0.00	$80.00
Calibration and certification	180	1,440	92.0	12.0	$0.00	$360.00	$552.00
Main controller comparison	300	4,740	184.2	24.0	$0.00	$0.00	$384.00
Component ablations	300	16,800	320.0	42.0	$0.00	$0.00	$672.00
Cross-renderer transfer	100	1,620	54.2	7.0	$0.00	$0.00	$112.00
Offline automated evaluation	300	—	26.0	3.5	$0.00	$0.00	$56.00
Human evaluation	150	—	—	—	$0.00	$900.00	$900.00
Optional API / audio extensions	50	420	—	—	$35.40	$0.00	$35.40
Failed runs and unplanned retries	—	84	8.5	1.2	$0.00	$0.00	$19.20
Storage, transfer, and other charges	—	—	—	—	$9.00	$0.00	$9.00
Listed phase-record sum	**1,460**	**25,784**	**737.4**	**96.7**	**$44.40**	**$1,260.00**	**$2,851.60**

Table 18: Arithmetic ledger of retained phase records, not a verified complete project invoice. Phase cases sum to 1,460 and may reuse the same cases. A unique-case total cannot be established from this table. Video and audio sum to $35.40, with $9.00 storage counted once in its own row. The listed total is $2,851.60; method coverage and invoice status require manifest records.

Controller	GPU-hours / case	P50 latency	P95 latency	Calls / case	Cost / case	Cost / successful case
Direct	0.14	162s	175s	4.0	$0.28	$2.33
Best-of-K	0.42	486s	512s	12.0	$0.84	$4.35
Single agent	0.31	380s	560s	8.6	$0.62	$2.41
Global repair	0.38	445s	680s	10.4	$0.76	$1.76
CONTRACT2VIDEO	0.28	330s	520s	7.9	$0.56	$1.04

Table 19: Retained inference profile with arithmetic cost checks, pending verification. Printed cost per case equals GPU-hours per case times the implied $2 rate; successful-case costs use the rounded Independent All rate. The case-versus-seed accounting and failed-call inclusion require a run manifest. The successful-case cost is undefined when no case succeeds.

Workflow	Evaluation condition	HCS	All	Human win	GPU-hours / case
VideoDirectorGPT	Shared-renderer adaptation	62.4%	26.8%	29.5%	0.32
VideoStudio	Original-system diagnostic	58.1%	21.0%	24.0%	0.35
VideoGen-of-Thought	Shared-renderer adaptation	66.8%	33.5%	36.0%	0.34
MovieAgent	Shared-renderer adaptation	69.2%	38.0%	39.5%	0.36
CONTRACT2VIDEO	Shared renderer and budget	78.6%	54.0%	—	0.28

Table 20: Published-workflow comparison. An adaptation is explicitly identified and is not presented as the original authors' reported system. Incompatible or unavailable systems are reported with reasons rather than replaced by a weak surrogate bearing the same name.

Setting	Controller	HCS	All	Calls / case	Cost / case	Failures
Video API	Direct	52.4%	16.0%	4.0	$0.32	0.0%
Video API	Global repair	74.0%	48.0%	10.2	$0.82	2.0%
Video API	CONTRACT2VIDEO	81.2%	58.0%	7.8	$0.62	0.0%
Audio extension	Without audio specialist	64.2%	31.0%	8.2	$0.66	2.0%
Audio extension	CONTRACT2VIDEO audio specialist	79.5%	52.0%	8.4	$0.67	0.0%

Table 21: Retained optional-extension summaries, pending verification. Cost/case is the video-call subtotal from printed calls at $0.08, rounded to two decimals; it excludes the $1.80 audio charge, whose per-case allocation requires a manifest. The visual and audio constraint inventories must not be pooled.

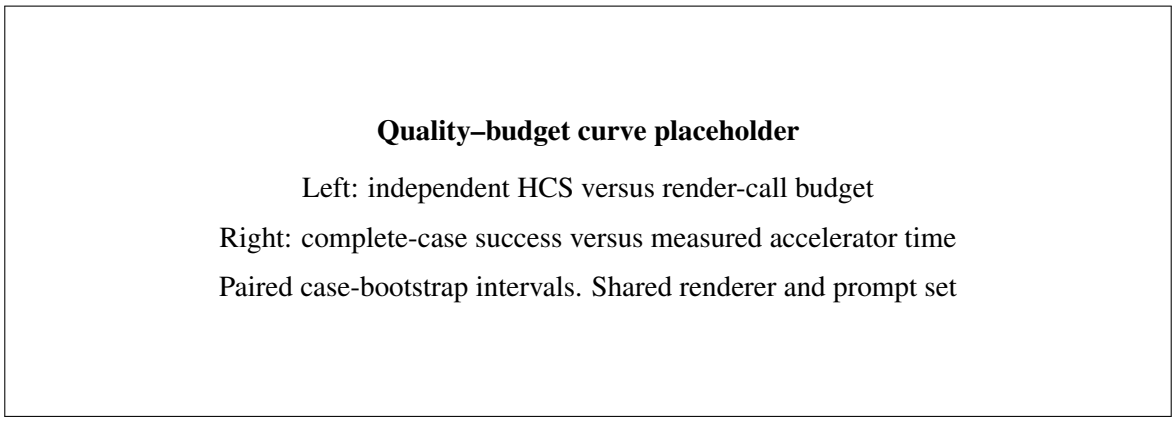


Figure 2: Planned quality–budget analysis. Report the initial four-shot output and subsequent checkpoints up to the twelve-attempt ceiling. The figure will expose whether any gain is explained by additional rendering.

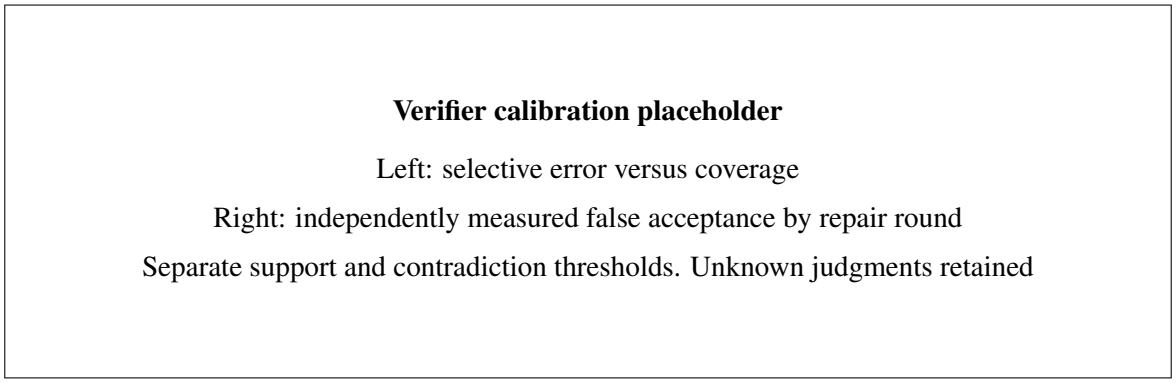


Figure 3: Planned calibration diagnostic. Fixed-split certification and trajectory-level empirical error are shown separately. The curve cannot substitute for a formal guarantee under adaptive distribution shift.

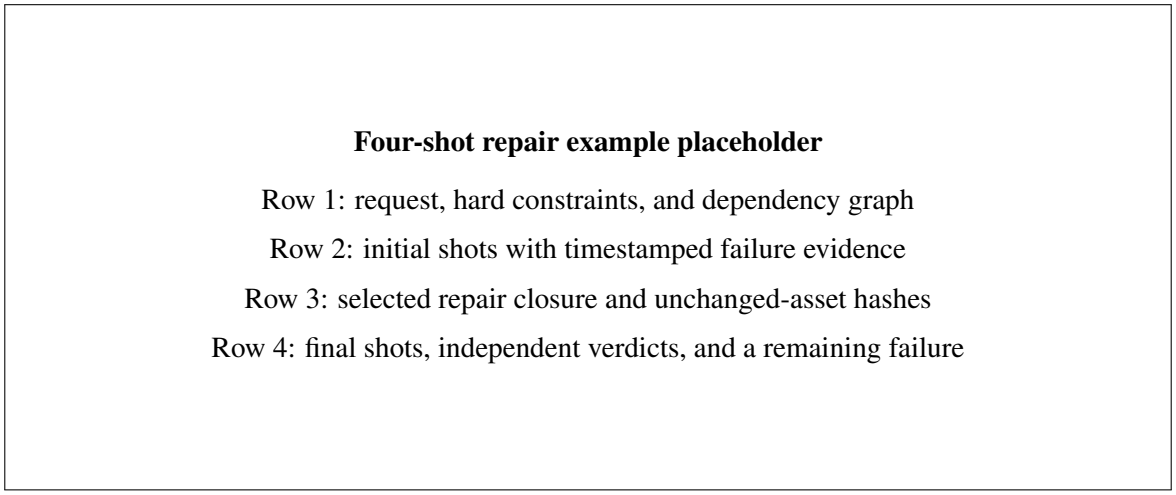


Figure 4: Planned qualitative comparison of local and global repair. The same prompts, seeds, and render-call budget are displayed. Unchanged assets are identified by hashes. Perceptual or semantic correctness requires independent judgments.